

Sheet 4 — Determinants

Numbering continues from Sheet 3.

66. Compute the determinants:

$$\begin{array}{llll} \text{a)} \begin{vmatrix} 2 & 3 \\ 2 & 6 \end{vmatrix} & \text{b)} \begin{vmatrix} 1 & 0 & 3 \\ 1 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix} & \text{c)} \begin{vmatrix} 4 & -8 & 9 \\ 0 & 0 & 0 \\ 1 & 12 & 8 \end{vmatrix} & \text{d)} \begin{vmatrix} a & x & aw \\ b & y & bw \\ c & z & cw \end{vmatrix} \end{array}$$

$$\begin{array}{ll} \text{e)} \begin{vmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ -2 & 0 & -3 \end{vmatrix} & \text{f)} \begin{vmatrix} 0 & 0 & 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & -1 & 2 & 2 & 5 \\ 0 & 3 & 2 & 5 & 5 & 6 \\ 2 & 2 & -2 & 1 & 1 & 1 \end{vmatrix} \end{array}$$

$$\begin{array}{ll} \text{g)} \begin{vmatrix} 2a & 2b & 2c \\ x & y & z \\ a+4x & b+4y & c+4z \end{vmatrix} & \text{h)} \begin{vmatrix} a & b & c \\ 0 & y & z \\ 2a & 2b+3y & 2c+z \end{vmatrix} \end{array}$$

$$\begin{array}{ll} \text{i)} \begin{vmatrix} 3 & -6 & 12 & 15 \\ 1 & 0 & 4 & 0 \\ 0 & 0 & -4 & 1 \\ 4 & 2 & 8 & 0 \end{vmatrix} & \text{j)} \begin{vmatrix} -4 & -2 & 3 & 1 \\ -3 & -1 & 4 & 2 \\ -5 & 3 & -4 & 2 \\ 6 & -4 & 5 & 3 \end{vmatrix} \end{array}$$

67. Find the values of x for which the following matrices are singular:

$$\begin{array}{ll} \text{a)} A = \begin{bmatrix} 1 & 2 & x \\ x & x+1 & x \\ 3 & 7-x & 2 \end{bmatrix} & \text{b)} B = \begin{bmatrix} x & 1 & 1 & 1 & 1 \\ 1 & x & 1 & 1 & 1 \\ 1 & 1 & x & 1 & 1 \\ 1 & 1 & 1 & x & 1 \\ 1 & 1 & 1 & 1 & x \end{bmatrix} \end{array}$$

68. Given that $\begin{vmatrix} a & b & c \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{vmatrix} = 1$, compute the determinant of the following matrices:

$$\begin{array}{ll} \text{a)} \begin{bmatrix} a & b & c \\ 6 & 3 & 0 \\ -\frac{1}{2} & -1 & -\frac{1}{2} \end{bmatrix} & \text{b)} \begin{bmatrix} a & b & c \\ 2a+2 & 2b+1 & 2c \\ a+1 & b+2 & c+1 \end{bmatrix} \\ \text{c)} \begin{bmatrix} a-1 & b-2 & c-1 \\ 3 & 3 & 1 \\ 1 & 2 & 1 \end{bmatrix} & \text{d)} \begin{bmatrix} 1 & -1 & -1 \\ 2 & 1 & 0 \\ 3a+1 & 3b+2 & 3c+1 \end{bmatrix} \end{array}$$

69. Without computing the determinants, show that:

$$\begin{array}{ll} \text{a)} \begin{vmatrix} a+b & b & 3c \\ x+y & y & 3z \\ 3 & 2 & 9 \end{vmatrix} = \begin{vmatrix} a+2x & x & 3 \\ b+2y & y & 6 \\ c+2z & z & 9 \end{vmatrix} & \text{b)} \begin{vmatrix} x+a & 3a & u+x+a \\ x+b & 3b & v+x+b \\ x+c & 3c & w+x+c \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ ax & bx & cx \\ 3u & 3v & 3w \end{vmatrix} \end{array}$$

70. Let A be a square matrix of order three such that $|A| = 4$. Compute the determinant of the matrices:

$$\begin{array}{llll} \text{a)} B = A^3 & \text{b)} 3A^{-1} & \text{c)} 2A^{-4} & \text{d)} } \frac{A^T A^{-1} A^3}{4} \end{array}$$

$$\text{e)} } A^2 \begin{bmatrix} 0 & 0 & 3 \\ 0 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}^{-1} A^T \frac{A^T}{2}$$

71. Compute the determinants:

$$\text{a) } \begin{vmatrix} 3 & 5 & 7 & 2 \\ 2 & 4 & 1 & 1 \\ -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 4 \end{vmatrix} \quad \text{b) } \begin{vmatrix} 1 & -2 & 3 & -4 \\ 2 & -1 & 4 & -3 \\ 2 & 3 & -4 & -5 \\ 3 & -4 & 5 & 6 \end{vmatrix} \quad \text{c) } \begin{vmatrix} 0 & 1 & 0 & 3 \\ 1 & 5 & 1 & 7 \\ 1 & 3 & 2 & 9 \\ 2 & 4 & 3 & 6 \end{vmatrix}$$

72. Find the values of x for which the following matrices are non-singular:

$$\text{a) } \begin{bmatrix} x+1 & x^2-x & 0 \\ 1 & 2 & 3 \\ x-1 & 2x & 0 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 2 & 0 & 1 & -1 \\ 1 & x & 0 & 1 \\ 0 & -1 & -1 & 2 \\ 1 & x & 0 & x \end{bmatrix}$$

73. Using the adjugate, find the inverse of the following matrices:

$$\text{a) } \begin{bmatrix} 2 & 5 \\ 4 & 7 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 1 & 1 & 2 \\ 4 & 5 & -1 \\ 2 & 0 & 1 \end{bmatrix} \quad \text{c) } \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

74. Given that A is a square matrix of order 3 and $\det A = 2$, compute $\det(\operatorname{adj} A)$.

75. Let A and B be square matrices of order n , with A invertible, such that $A = A^T$ and $B^{-1} = B^T$.

a) Show that $|B|^2 = 1$.

b) Show that the matrix equation $B(A^{-1}X)^T A = |B| B \operatorname{adj} B + B$ has the solution $X = B + I$.

76. Given that A is a non-singular matrix of order n , compute:

a) $\det(\operatorname{adj} A)$ and $\det(\operatorname{adj}(\operatorname{adj} A))$ in terms of $\det A$ and n .

b) Given that B is a real matrix of order 3 and that $\det B = 4$, compute the value of:

i) $\det(-B)$ ii) $\det(10B)$ iii) $\det(B^2)$ iv) $\det(B^{-1})$ v) $\det(\operatorname{adj} B)$ vi) $\det(\operatorname{adj}(B^{-1}))$
vii) $\det((\operatorname{adj}(\operatorname{adj} B))^{-1})$

77. Using the theory of determinants, show that the following systems are consistent and solve them:

$$\text{a) } \begin{cases} 3x + 2y = 5 \\ 2x - y = 8 \end{cases} \quad \text{b) } \begin{cases} x + y - z = 0 \\ 2x + y - 2z = -2 \\ 2x + 3y = 8 \end{cases}$$

78. Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Show that the system of equations $AX = B$ has a unique solution and solve it using Cramer's rule.

79. Find the value of z that satisfies the following system of equations:

$$\begin{cases} x + 2y + 2w = 1 \\ x + 2y + z + w = 1 \\ 2x + 3y + 6w = 1 \\ 2y + z + w = 1 \end{cases}$$

80. Solve the following system using Cramer's rule:

$$\begin{cases} 2x_1 + x_2 + 5x_3 + x_4 = 5 \\ x_1 + x_2 - 3x_3 - 4x_4 = -1 \\ 3x_1 + 6x_2 - 2x_3 + x_4 = 8 \\ 2x_1 + 2x_2 + 2x_3 - 3x_4 = 2 \end{cases}$$

81. Use determinants to find the values of k for which the following system has a unique solution:

$$\begin{cases} kx + y + z = 1 \\ x + ky + z = 1 \\ x + y + kz = 1 \end{cases}$$

82. Find α such that $x = 3$:

$$\begin{cases} 3y + 2x = z + \alpha \\ 3x + 2z = 8 - 5y \\ 3z - 1 = x - 2y \end{cases}$$